



2. Strong interaction



STARTSEITE

EINFÜGEN

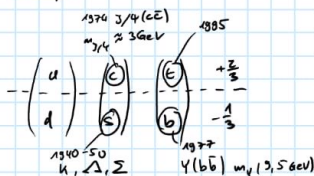
ZEICHNEN

ANSICHT



Beenden

2) Quarks and colour

Family of quarksspin- $\frac{1}{2}$ particlesmesons $|q\bar{q}\rangle$ $|\bar{q}q'\rangle$ different quarks
of same or different typebaryons $|qq'q''\rangle$ $|\bar{q}\bar{q}'\bar{q}''\rangle$ colourall up quarks in spin-up state1951 Fermi $\Delta^{++} = |u u u\rangle$ $J_\Delta = \frac{3}{2}$

forbidden by Fermi statistics (fermions must have different quantum numbers)

 $\Rightarrow$ we need another quantum number: colour charge3 colours R, G, B and 3 anti colours $\bar{R}, \bar{G}, \bar{B}$

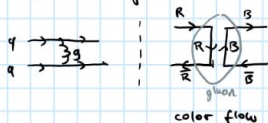
hadrons are 'colourless'

 $|RR\rangle, |GG\rangle, |BB\rangle$ $|RGB\rangle$ + combinationsgluons

- mediator of the strong interaction

 $m_g = 0$

- bicolour objects



- colour-anticolour state

3 possibilities